Persistent Information in Large Agent Populations

Scaling, Incentives, and Coordination

Research summary accompanying the integrated working paper

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The connected question

How do the scaling of information deposits and the incentives to produce them jointly determine the information environment that survives in a large population? Exogenous production establishes the possible limits; the game asks which one decentralized incentives sustain. A productive coordinated policy shows that disappearance can coexist with a strictly useful alternative. Creator credit examines one response and its limitations.

One common model

Agent i has state $Y_i = (X_i, C_i) \in \{0,1\} \times \{U, E, A\}$: a binary task coordinate and an unexposed, exposed, or adopter communication state. The population law is $\mu^N = N^{-1} \sum_i \delta_{Y_i}$, and $a(\mu)$ is its adopter share. A shared board $B \geq 0$ decays at rate κB. Communication follows

$$U \xrightarrow{\rho B} E, \qquad E \xrightarrow{\alpha} A, \qquad E \xrightarrow{\nu} U, \qquad A \xrightarrow{\delta} U.$$

Tasks complete at rate c_0 for nonadopters and c_1 for adopters, and reset at rate r. Deposits change only the board. The game observes labeled states and the board, and controls only artifact production. [Paper, §2]

Exogenous channel	Per-adopter rate	Board jump	Limiting effect
Routine	λ_w	η/N	Deterministic drift
Artifact	λ_M/N	h	State-thinned common jumps
Hybrid	Both channels	Both sizes	Drift plus common jumps

These rates are prescribed, not equilibrium choices. In the game, an eligible player's artifact rate is $\lambda_M u_i/N$, with $0 \leq u_i \leq \bar{u}$. Frozen unit production is feasible as a game policy only when $\bar{u} \geq 1$.

What the probability foundation establishes

Writing $Q(B)$ for the common transition matrix, the frozen hybrid limit is

$$\dot{\mu}_t = \mu_t Q(B_t), \qquad dB_t = [-\kappa B_t + \eta \lambda_w a(\mu_t)]dt + h\, dJ_t,$$

where J accepts a common Poisson proposal at height at most $\lambda_M a(\mu_{t-})$. The weak and rare limits disable the corresponding channel. The board jumps; the population law remains continuous.

On each fixed horizon, the expected uniform error in population, board, and accepted count is at most $C_T(\varepsilon_{N,p} + N^{-1/2})$ for these frozen models. Assumptions include bounded initial boards, L^p convergence to deterministic initial data for some $p > 2$, and independence from future clocks. The proof includes the random finite hybrid board explicitly. [Paper, Thms. 3.2–3.4; App. A]

With exchangeable chaotic initialization, weak-limit tagged paths have an ordinary product law. Rare and hybrid tags factor conditional on the common proposal measure. Averaging over that history gives a mixture of products, so unconditional dependence may remain. This is not finite-N independence.

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Incentives select a limiting environment

The allocation-and-readiness institution. Opportunities arrive at total rate $\lambda_M \bar{u}$ and are allocated uniformly among N players. An eligible recipient succeeds with probability $u_i/\bar{u}$. Ineligible assignments expire and cannot be reassigned. Players choose and pay for readiness before allocation, whether an opportunity arrives or not. Readiness units, cap, costs, and per-person rewards stay fixed across N. Fixed capacity alone does not imply these restrictions. [Paper, §5]

Only adopters choose effort, with cost $c(u) = \ell u + \gamma u^2/2$, $\ell \geq 0$, $\gamma > 0$. On $[0, T]$, players maximize expected discounted reward minus cost, with $0 \leq g \leq R$, $\beta \geq 0$, and zero terminal payoff. Welfare is the per-person average. There are initially no transfers.

An equilibrium exists before taking the limit. For every finite N, T, the paper constructs a pure continuous Markov-perfect equilibrium and verifies it against admissible predictable unilateral deviations. Uniqueness is confined to the bounded jointly continuous mild-HJB class. Bounded board jumps perturb the artifact-disabled semigroup; verified reward bounds continue the local construction across the horizon. The potential/interlacing proof is included. [Paper, Thm. 5.1; App. B]

Private incentives suppress artifacts. Let $\Delta_h V_i = V_i(s^h) - V_i(s)$ and $K_T = R \int_0^T e^{-\beta t} dt$. The value bounds $0 \leq V_i \leq K_T$ give

$$u_i^{\mathrm{NE}} = \mathrm{clip}_{[0,\bar{u}]} \left(\frac{(\lambda_M/N)\Delta_h V_i - \ell}{\gamma} \right) \leq \frac{[\lambda_M K_T/N - \ell]_+}{\gamma}.$$

Thus, for quadratic cost,

$$\Lambda_N^{\mathrm{NE}} = \frac{\lambda_M}{N} \sum_i \mathbf{1}_{\{C_i = A\}} u_i^{\mathrm{NE}} \leq \frac{\lambda_M^2 K_T}{\gamma N}.$$

Positive ℓ gives exact shutdown for sufficiently large N. The value difference already includes future task benefits and rivals' feedback; the $1/N$ factor is in individual discovery technology. [Paper, Cor. 5.2]

Equilibrium and artifact-disabled paths can agree until the first artifact, whose fixed-horizon probability is $O(N^{-1})$. Their expected uniform state and welfare differences are $O(N^{-1})$ with bounded initial boards. Combining this with the weak foundation gives a deterministic equilibrium limit, possibly with routine drift, and ordinary tagged independence. [Paper, Prop. 7.1]

Disappearance can coexist with productive coordination

Take rare-only production, $B_0 = 0$, reward $g(x, c) = Rx$, positive $R, \rho, \alpha, h, \lambda_M, \bar{u}$, and $c_1 > c_0$. Uniformly positive fractions initially are adopters and incomplete unexposed players. A sufficiently small fixed effort $\varepsilon > 0$ by adopters until $t_0 < T$ yields

$$W_N^{\varepsilon} - W_N^0 = \varepsilon A_N + O(\varepsilon^2), \quad A_N \geq A_* > 0, \qquad \liminf_N (W_N^{\mathrm{SP}} - W_N^{\mathrm{NE}}) \geq \delta_* > 0.$$

The constructive theorem is restricted to this seeded, zero-board, rare-only region. Small positive linear costs can also be allowed; arbitrary initial boards and hybrid productivity require a separate positivity argument. [Paper, Thms. 6.1, 6.3]

The same specified policy has a common-jump limit. For two limiting tagged discounted task rewards H_i, conditional independence and a productive early-artifact event give $\mathrm{Cov}(H_1, H_2) = \mathrm{Var}(\mathbb{E}[H_1 \mid \Pi^0]) > 0$. This shows that the common history affects useful outcomes, not that randomness itself raises welfare. For general bounded spatially Lipschitz feedback, the paper gives a quantitative rare-only limit and a qualitative hybrid limit. Neither optimal-value nor optimal-control convergence is established. [Paper, Thm. 7.2; §7]

Two scaling questions, and a benchmark response

Physical deposition scaling. For an order-one eligible share, individual rate $\lambda_0 N^{-\zeta}$ and positive impact $\eta_0 N^{-\xi}$ give aggregate rate, drift, and predictable quadratic-variation orders

$$N^{1-\zeta}, \qquad N^{1-\zeta-\xi}, \qquad N^{1-\zeta-2\xi}.$$

For $\zeta, \xi \geq 0$, order-one positive drift lies on $\zeta+\xi = 1$. Positive ξ gives vanishing jumps and fluctuations; $(1, 0)$ is the finite-jump endpoint. The accepted rate can be state dependent. This is uncentered positive-deposit accounting, not a Brownian-limit theorem or an incentive classification. [Paper, Prop. 3.1]

Strategic productivity scaling. The two-state benchmark instead has absent/present artifact state Z, benefit vZ, depreciation rate d, infinite horizon, and fixed $\beta > 0$. When absent, discovery is $a_N \sum_i u_i$. With quadratic cost, $a_N = \lambda N^{-s}$, fixed positive $\lambda, v, \gamma, \bar{u}$, and $k = \beta + d$, the symmetric stationary equilibrium's absent-state hazard satisfies

$$q_N^{\mathrm{NE}} \sim \begin{cases} \lambda\sqrt{v/\gamma}\, N^{1/2-s}, & 0 \leq s < 1/2, \\ \lambda^2 D_c/\gamma, & s = 1/2, \\ \lambda^2 v(\gamma k)^{-1} N^{1-2s}, & s > 1/2, \end{cases} \qquad kD_c + \lambda^2 D_c^2/\gamma = v.$$

Every fixed positive cap is eventually nonbinding. Positive linear cost instead shuts down every $s > 0$ eventually. At $s = 0$, activity requires $\lambda v > k\ell$, with limiting hazard $\lambda v/\ell - k$. These are benchmark results, not six-state scaling theorems. Stationary event frequency is $dq_N/(d + q_N)$ when $d > 0$, rather than the conditional hazard q_N. No exchange of population and long-horizon limits is asserted. [Paper, Prop. 4.3; §4]

With normalized productivity $a_N = \lambda/N$, Nash artifact intensity vanishes but an active benchmark planner sustains a positive constant hazard. The capped and linear-cost cases, waiting-time laws, occupancy, and values are solved in the paper. Concentrating the same opportunity stream on one designated creator also gives an N-independent private problem. Capacity alone therefore cannot explain collapse. This comparison does not derive endogenous hierarchy or efficient delegation. [Paper, Prop. 4.2]

Creator credit: exact implementation and a counterexample

Pay the successful creator Np and charge every player p per discovery, including the creator. Transfers balance event by event. In the two-state benchmark, $p = D^{\mathrm{SP}}$, the planner's value gap, implements planner effort as a symmetric stationary Markov equilibrium. The assumptions include risk neutrality, reliable attribution, enforceable levies, and commitment. [Paper, Prop. 8.1]

For an interior six-state planner target, the actual individual continuation value under transfers, U_i^p, must instead satisfy

$$\Delta_h U_i^p + (N-1)p = N\Delta_h W_N^{\mathrm{SP}}.$$

Budget balance gives only an average value identity. Setting $p = \Delta_h W_N^{\mathrm{SP}}$ need not satisfy this individual condition. In a two-agent example, a completed adopter has $\Delta_h U_1^p = 0$; halving prescribed effort over a positive interval, then reverting, gives a strictly positive gain. This proves nonimplementation, not a full optimal dynamic best-response policy. [Paper, Prop. 8.2; §8]

Under active normalized benchmark implementation and positive expected discovery, a tagged player's gross creator payments tend to zero in probability but keep a positive expectation: they are not uniformly integrable. State-path convergence alone therefore cannot pass expected transfer payoffs to a limit. [Paper, §8]

Numerical evidence and a secondary application

The illustrated six-state policy uses eligible effort 0.25 until time 1.25 on a horizon of 6, with zero board, no routine deposits, 20% initial adopters, and all tasks incomplete. Each cell uses 160 independent histories; the paper specifies the full configuration and methods. [Paper, §10]

Population	Welfare above exact baseline (95% interval)
60	0.1147 $[0.0469, 0.1825]$
240	0.1947 $[0.1184, 0.2710]$
960	0.1240 $[0.0618, 0.1863]$
Fixed-policy limit	0.0998 $[0.0442, 0.1554]$

These are per-capita discounted net rewards with Student intervals; the exact baseline is 1.15065544. The conservative witness certifies effort about 3.70×10^{-7} and gain about 1.78×10^{-10}. The larger improvement at 0.25 is simulation evidence. An imposed $0.25/N$ diagnostic is not a computed Nash equilibrium. Histories were retained, with aggregates rechecked; tests and deterministic calculations were rerun. Code remains local.

The weak limit also closes on exposed share, adopter share, and board stock. Its loop gain is

$$G_B = \frac{\rho\alpha\eta\lambda_w}{(\alpha+\nu)\delta\kappa}.$$

Under the stated positive-rate assumptions, $G_B = 1$ is a forward transcritical threshold with local full-simplex stability exchange. No global full-system stability is asserted. This secondary reservoir-style result is distinct from the production game. [Paper, Prop. C.1]

Interpretation and remaining boundaries

The integrated paper includes the common definitions and central proofs. Its access institution is unvalidated for AI-agent systems. Known productivity is not Bayesian learning; endogenous hierarchy, efficient specialization, a general heterogeneous-state mechanism, and optimizer convergence are not established. The scalar board omits content and artifact-specific exposure.

A next question is how reassigning unused opportunities changes the comparison at fixed capacity and task dynamics. This remains unsolved.

Related work

Team discovery and durable public goods exhibit incentive wedges [2, 1]; common-Poisson limits supply probability context [3]. The full paper discusses the closest comparisons without claiming priority.

Selected references

[1] Marco Battaglini, Salvatore Nunnari, and Thomas R. Palfrey. Dynamic free riding with irreversible investments. *American Economic Review*, 104(9):2858–2871, 2014. doi: 10.1257/aer.104.9.2858. URL https://snunnari.github.io/bnp_dynfreerider_aer.pdf.

[2] Alessandro Bonatti and Johannes Hörner. Collaborating. *American Economic Review*, 101(2):632–663, 2011. doi: 10.1257/aer.101.2.632. URL https://bpb-us-w2.wpmucdn.com/campuspress.yale.edu/dist/3/352/files/2012/02/65916.pdf.

[3] Daniel Hernández-Hernández and Joshué Helí Ricalde-Guerrero. Conditional McKean–Vlasov differential equations with common Poissonian noise: Propagation of chaos. *Stochastics and Dynamics*, 25(7–8):2550033, 2025. doi: 10.1142/S0219493725500339. URL https://arxiv.org/abs/2308.11564. Article 2550033; author version arXiv:2308.11564v3.